**PRESENTED BY**
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# MACHINE LEARNING BASICS.

DAY 2/30 | AI
INTERVIEW PREP

**Top 10 ML Concepts Frequently Asked in AI & Machine Learning Interviews**

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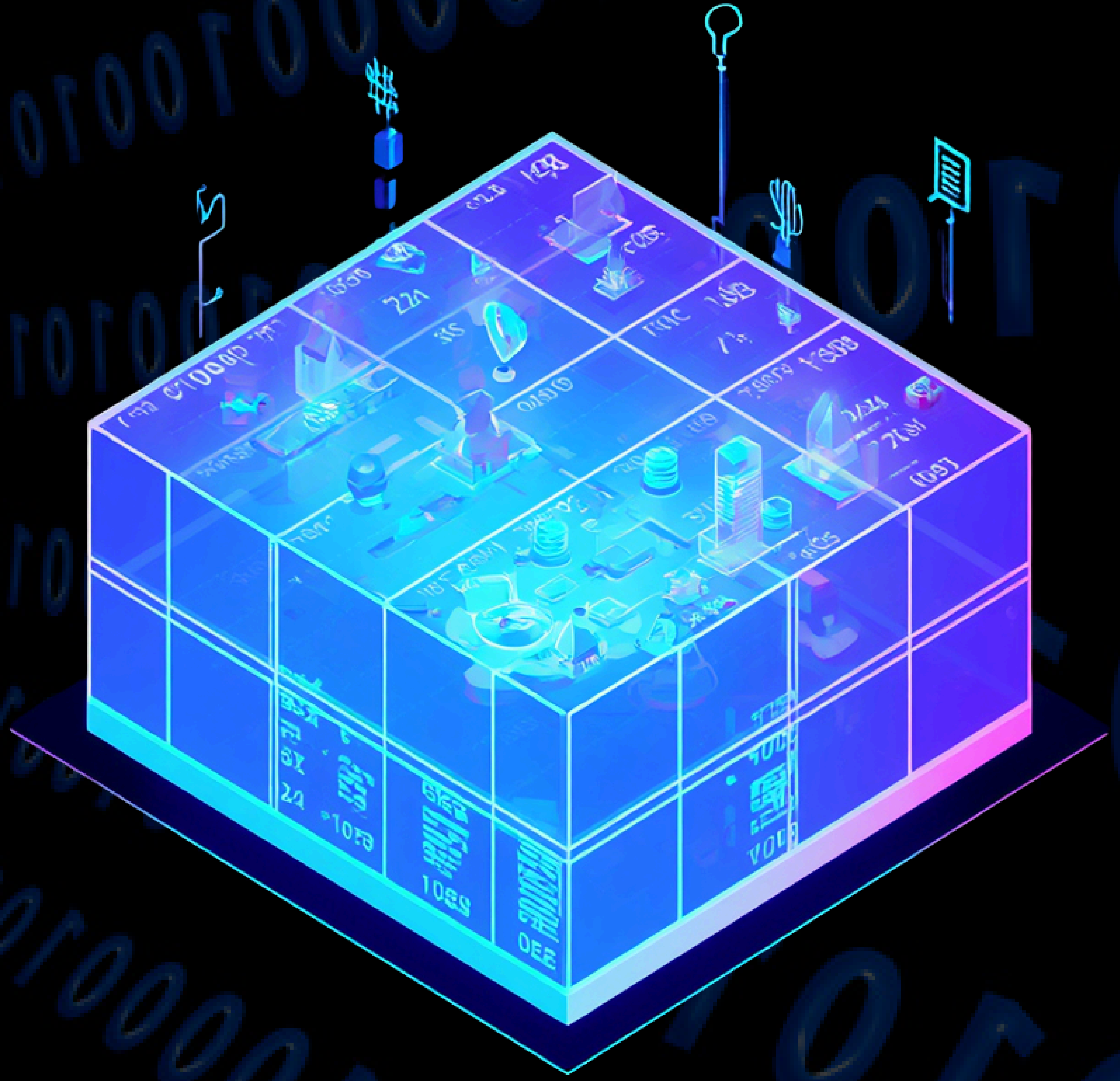
# What is a Dataset?

A dataset is a collection of data used to train, validate, and test a machine learning model.

📌 Real-World Example:
A customer database with columns like Age, Income, and Purchase History — each row is one data point.

🎯 Key Interview Points:

- A dataset is the foundation of every machine learning model
- Datasets can be structured (CSV, Excel, SQL tables)
- Datasets can be unstructured (Images, Audio, Text)
- Better data quality usually leads to better model performance



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# Features in Machine Learning

Features are the input variables or attributes used by a machine learning model to make predictions.

📌 Real-World Example:
For a house price prediction model, features may include the size of the house, number of bedrooms, location, age of the property, and number of bathrooms. These inputs help the model estimate the house price.

🎯 Key Interview Points:

- Features are the information provided to the model
- Also known as Independent Variables or Predictors
- Each feature represents a measurable property of the data
- Better features often lead to better model performance

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# Labels – Target Output Variables

Labels are the outputs that a machine learning model learns to predict. They represent the correct answer associated with the input features.

📌 Real-World Example:
In a house price prediction model, the house price is the label. The model uses features such as size, location, and number of bedrooms to predict this value.

🎯 Key Interview Points

- Labels are also known as Target Variables
- In supervised learning, every training example has a corresponding label
- The model learns the relationship between features and labels
- Labels represent the desired output of the model

📚 Types of Labels

- Continuous Labels → Regression (Price, Temperature, Salary)
- Categorical Labels → Classification (Spam/Not Spam, Cat/Dog)

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# Features vs Labels

## Features

- Input variables provided to the model
- Also called Predictors or Independent Variables
- Describe the characteristics of the data
- Used to make predictions
- Example: Size, Location, Bedrooms

## Labels

- Output the model tries to predict
- Also called Target Variables
- Represent the correct answer
- Used to evaluate predictions
- Example: House Price

## 🎯 Key Interview Points

- Features are the inputs to a machine learning model
- Labels are the outputs the model learns to predict
- A model learns the relationship between features and labels
- Features = X, Labels = Y

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# Training Set

A training set is the portion of the dataset used to teach a machine learning model. The model learns patterns, relationships, and rules from this data during training.

📌 Real-World Example:
Imagine teaching a child to identify cats using hundreds of labeled cat photos. By repeatedly seeing these examples, the child learns what a cat looks like. Similarly, a machine learning model learns from the training data.

🎯 Key Interview Points:

- The training set is used to learn patterns from data
- It is usually the largest portion of the dataset
- A common split is 70–80% Training Data
- The model should not be evaluated using only training data

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# Validation Set

A validation set is a portion of the dataset used to evaluate a model during training. It helps measure performance on unseen data and guides model improvement before final testing.

📌 Real-World Example:
A student takes practice tests before the final exam to identify weaknesses and improve performance.

🎯 Key Interview Points:

- Typically 10–20% of the dataset
- Used during training, not for learning
- Helps detect overfitting
- Used to tune and select the best model
- Different from the Test Set

# Test Set

A test set is a portion of the dataset used to evaluate the final performance of a machine learning model. It represents completely unseen data and helps measure how well the model generalizes to real-world scenarios.

📌 Real-World Example:
Think of it as the final exam. The model learns using the training set, improves using the validation set, and is evaluated using the test set.

🎯 Key Interview Points:

- Typically 10–20% of the dataset
- Used only after training is complete
- Represents unseen, real-world data
- Used to measure the final performance of the model
- Should never be used during training or model tuning

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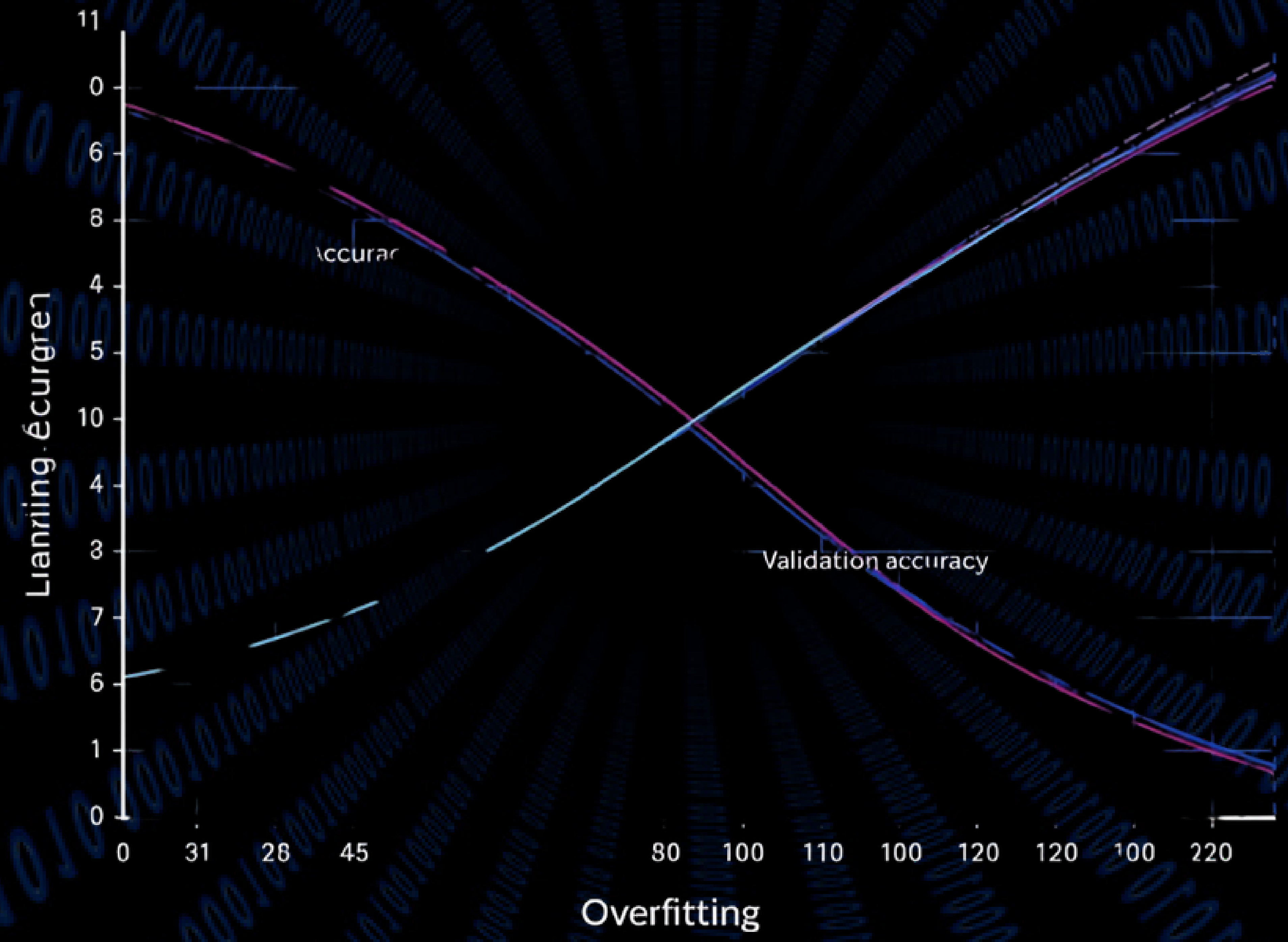


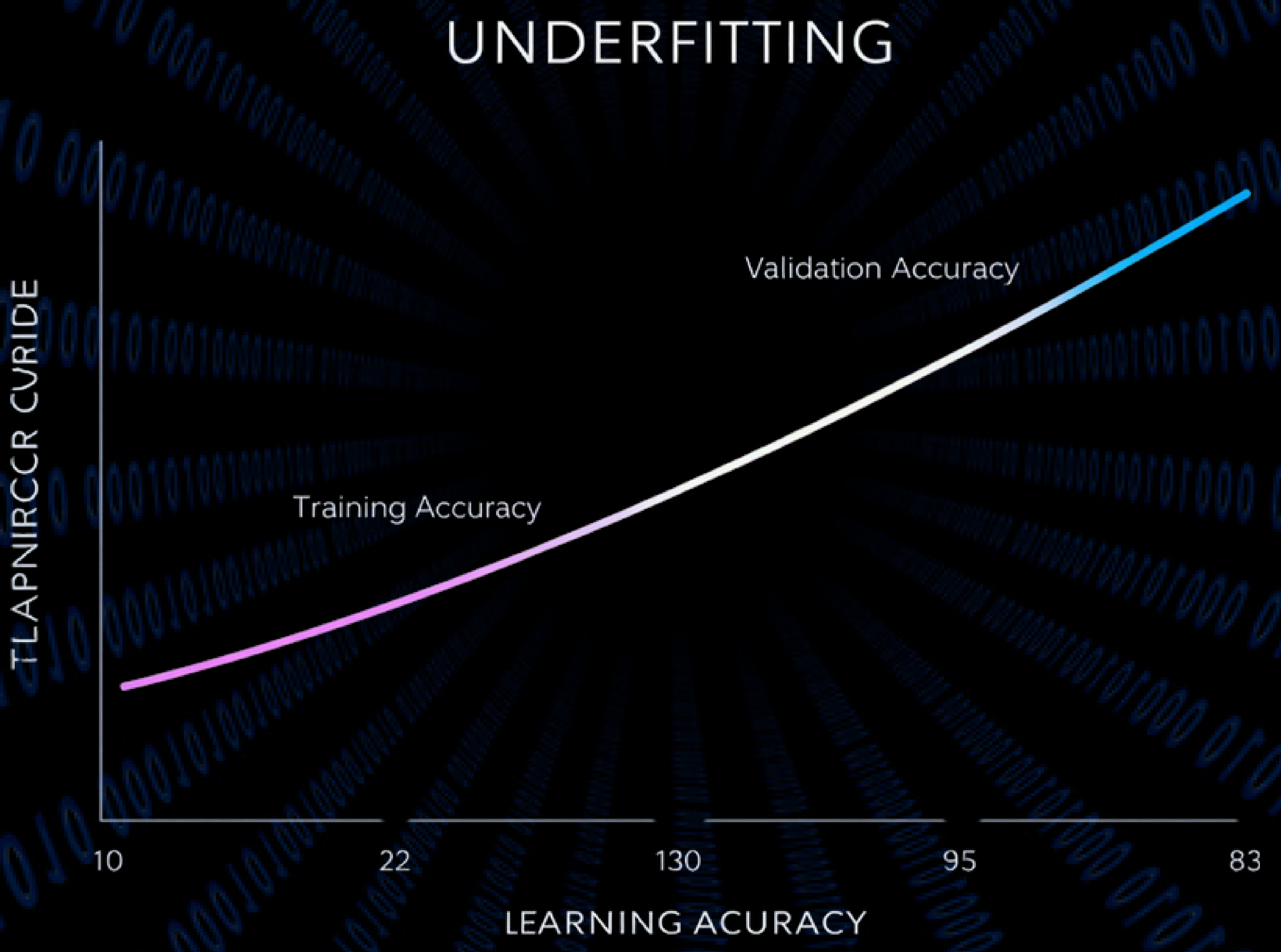

# Overfitting

Overfitting occurs when a machine learning model learns the training data too well, including its noise and patterns, but fails to generalize to new, unseen data.

📌 Real-World Example:
Imagine a student who memorizes all the answers from previous exams instead of understanding the concepts. They perform exceptionally well on familiar questions but struggle with new ones.

🎯 Key Interview Points:

- High Training Accuracy
- Low Test Accuracy
- Poor performance on unseen data
- Indicates that the model has memorized rather than learned
- Common in overly complex models




# Underfitting

Underfitting occurs when a machine learning model is too simple to learn the underlying patterns in the data. As a result, it performs poorly on both training and unseen data.

📌 Real-World Example:
Imagine trying to fit a straight line to data that actually follows a curve. The model fails to capture the true pattern and makes inaccurate predictions.

🎯 Key Interview Points:

- High Training Accuracy
- Low Test Accuracy
- Poor performance on unseen data
- Indicates that the model has memorized rather than learned
- Common in overly complex models

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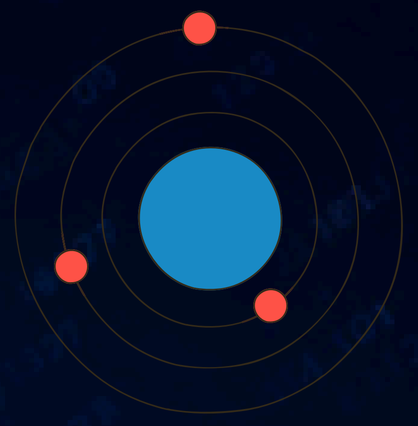
# Bias-Variance Tradeoff



### 🎯 High Bias (Underfitting)

The model is too simple to capture the underlying patterns in the data.

- High Training Error
- High Test Error
- Misses important patterns
- Often leads to Underfitting

### 🎯 Optimal Balance

The ideal point where the model generalizes well to unseen data.

- Low Training Error
- Low Test Error
- Good balance between Bias and Variance
- Goal of every Machine Learning model

### 🎯 High Variance (Overfitting)

The model is too complex and memorizes the training data instead of learning general patterns.

- Low Training Error
- High Test Error
- Poor performance on unseen data
- Often leads to Overfitting

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# Quick Revision

✔ Dataset – Collection of data used for training
✔ Features – Input variables (X)
✔ Labels – Output/target variable (Y)
✔ Features vs Labels – Inputs vs Outputs
✔ Training Set – Model learns from this
✔ Validation Set – Tune & evaluate model
✔ Test Set – Final unbiased evaluation
✔ Overfitting – Too fit to training data
✔ Underfitting – Model too simple
✔ Bias-Variance Tradeoff – Balance error types

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# THANK YOU!

FOR LEARNING WITH US TODAY

AI Interview Preparation – Machine Learning Basics